

DIABETES 30-DAY READMISSION ANALYSIS

Business Report | Python + SQL Analytics Project

Dataset: Diabetes hospital encounter dataset

Analytical focus: Observed 30-day hospital readmission patterns

Primary tools: Python (pandas, Matplotlib, Seaborn) + MySQL

This report follows the structure and business-report style of the supplied Sales Intelligence project report, while using only findings supported by the Project 3 notebook, SQL files, and project documentation.

1. Executive Summary

This project analyzes **101,766 diabetes-related hospital encounters** to understand which patient-history, hospital-utilization, clinical-complexity, and treatment-related characteristics are associated with different observed 30-day readmission rates. The analysis uses Python for data cleaning, exploratory analysis, segmentation, and focused visualization, and SQL for database validation and business-focused analytical queries.

The overall observed 30-day readmission rate is **11.16%** (11,357 readmissions). The strongest descriptive signal is previous healthcare utilization: encounters with 2+ previous inpatient visits have a **21.40%** observed readmission rate, while 2+ previous emergency visits have a **21.26%** rate. When the two histories are combined, the 2+ inpatient / 2+ emergency segment reaches **28.59%**.

Business Problem

Hospitals need to understand which encounter segments show higher observed readmission rates so that care-management, discharge planning, and follow-up resources can be better prioritized. Raw encounter records make these patterns difficult to see without systematic cleaning, grouping, and comparison.

Business Objectives

- Identify major patient and utilization segments with different observed 30-day readmission rates.
- Compare prior inpatient, emergency, and outpatient utilization.
- Examine hospital stay duration and clinical-complexity indicators such as medications, laboratory procedures, diagnoses, and procedures.
- Examine diabetes-related variables such as A1C result, maximum serum glucose, insulin, and diabetes medication.
- Reproduce important metrics directly in MySQL using SQL.
- Communicate only the strongest findings through focused Python visualizations rather than charting every column.

2. Dataset, Data Quality & Analytical Setup

The notebook contains **101,766 encounters** and **71,518 unique patients**. The analysis is encounter-level: one patient can have multiple encounters, so most rates describe the proportion of encounters that resulted in a 30-day readmission.

Data Quality Checks

- Full duplicate rows: **0**.
- Duplicate encounter IDs: **0**.
- The literal '?' placeholder was converted to pandas missing values.
- A1C and maximum-glucose missing categories were retained as **Not Measured**.
- Race and diagnosis fields with missing values were represented as **Unknown**.
- Medical specialty and payer fields with missing values were represented as **Not Recorded**.

Target Definition

The original readmission field contains three categories: NO, >30, and <30. The project creates **readmitted_30d**, where <30 is coded as 1 and all other categories as 0. This makes the outcome suitable for simple counting and rate calculations.

Core KPI

Metric	Value
Total encounters	101,766
Unique patients	71,518
30-day readmissions	11,357
Overall observed rate	11.16%

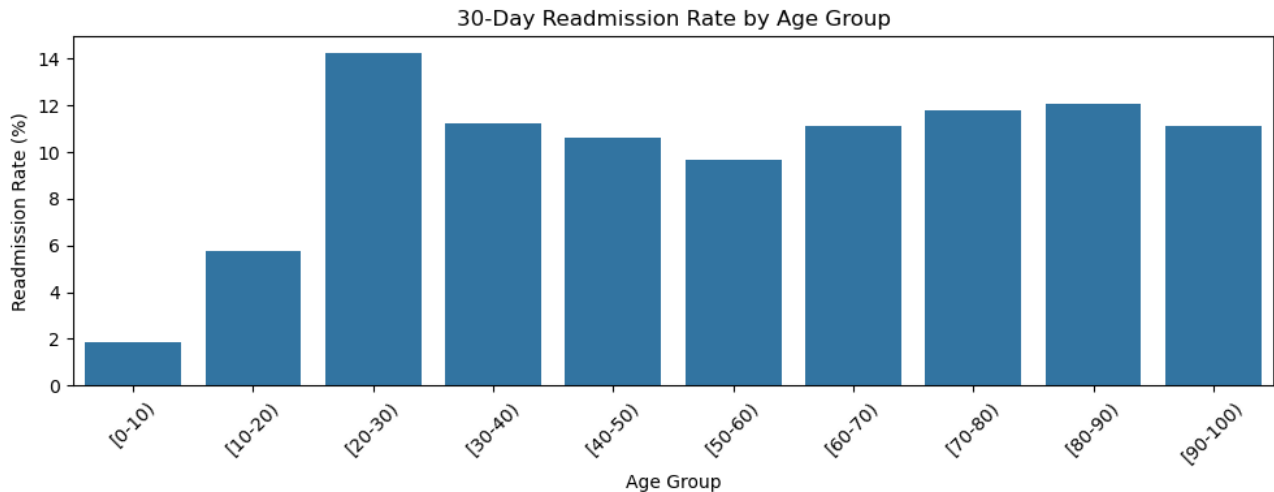
The project consistently uses the same rate definition across segments: readmissions divided by encounters, expressed as a percentage. This makes the comparisons across age, utilization, and clinical groups consistent.

3. Python EDA — Demographics

The demographic analysis was used to establish context before focusing on the stronger utilization signals. The notebook intentionally visualizes only selected findings that answer useful analytical questions.

Age

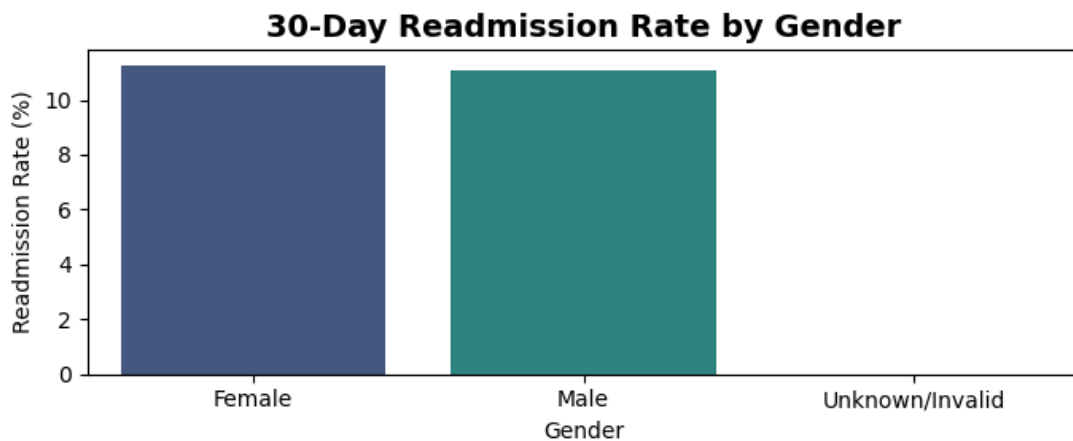
Observed readmission rates vary by age group. The highest rate in the age table is **14.24%** for the 20–30 group, but that group contains only 1,657 encounters. Larger older groups, especially 70–80 and 80–90, also sit above the 11.16% overall baseline.



Business interpretation: Age shows differences, but the smaller 20–30 group should not be treated as the main business signal. Denominator size matters.

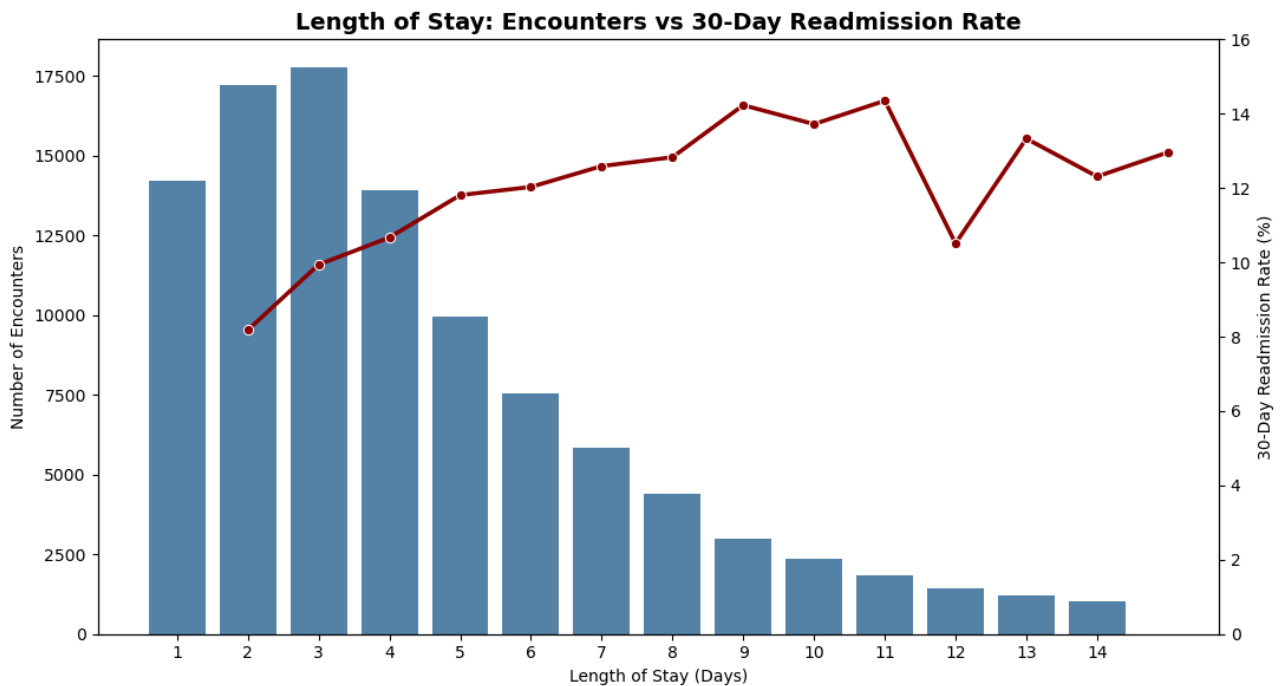
Gender

Female encounters show an 11.25% observed rate versus 11.06% for male encounters. The difference is very small, so gender is a supporting demographic observation rather than a headline finding.



4. Hospital Utilization & Clinical Complexity

Length of stay provides an important example of why both **rate** and **volume** should be examined together. Encounter volume falls considerably as hospital stay increases, while the observed readmission rate generally rises.



Key observation: The observed rate rises from 8.18% for a 1-day stay to 14.35% at 10 days, but the number of encounters falls from 14,208 to 2,342. A higher percentage does not automatically mean a larger absolute number of readmissions.

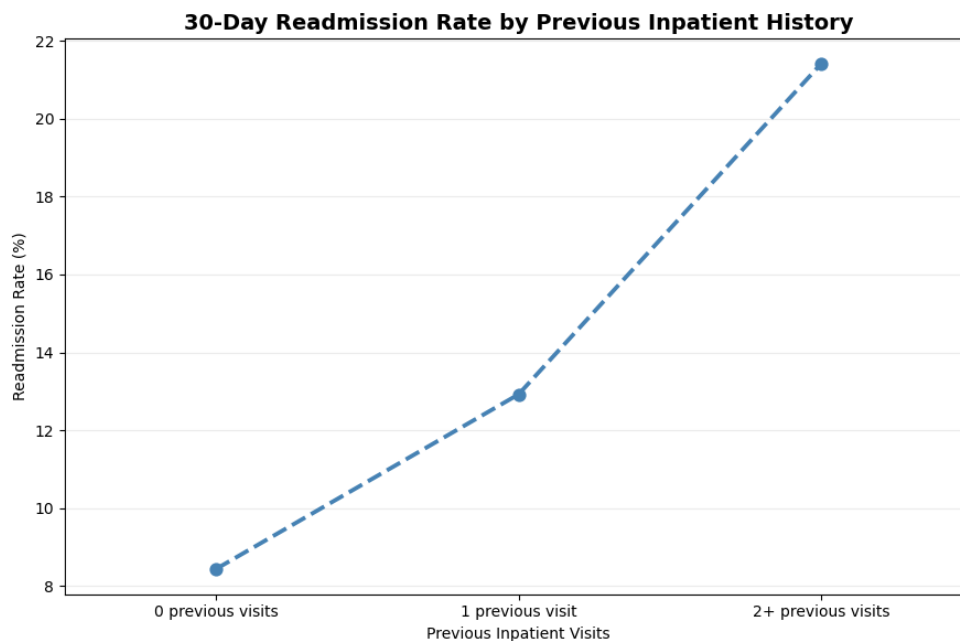
Medication and laboratory complexity

Readmission rates increase gradually with both laboratory-procedure groups and medication-count groups. Medication count is particularly clear: the observed rate increases from **9.06%** in the low group to **12.78%** in the very-high group.

Diagnosis complexity also shows an upward pattern from 8.49% in the low group to 12.39% in the high group. The 12+ diagnosis group is extremely small and should not be over-interpreted.

5. Prior Healthcare Utilization — Strongest Finding

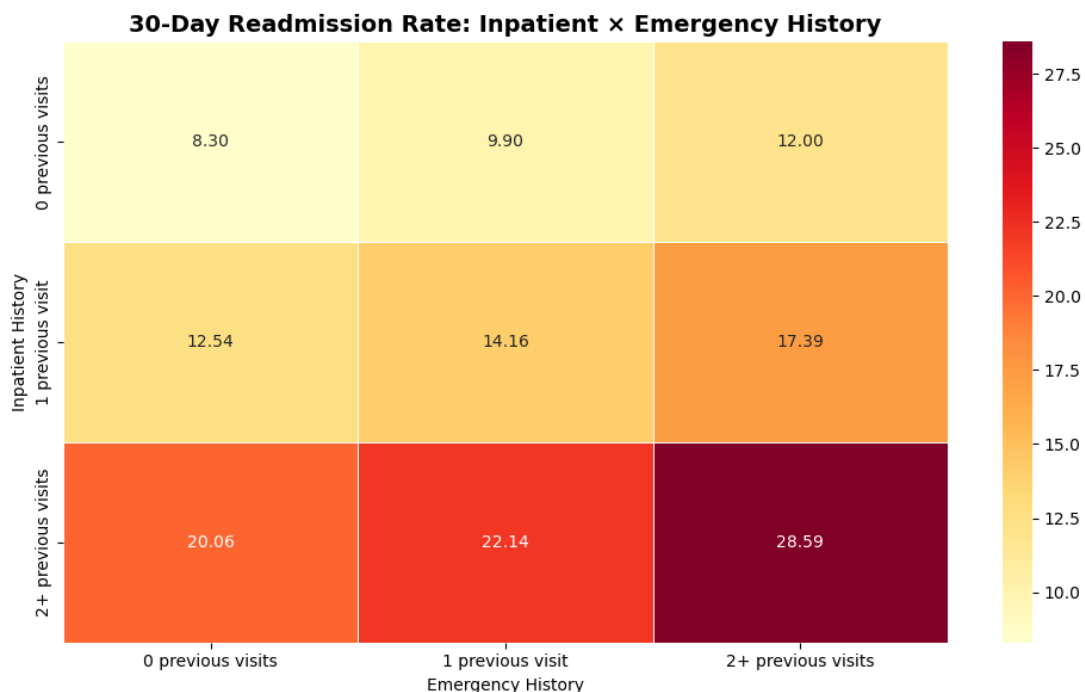
Previous inpatient and emergency utilization are the strongest descriptive signals in the project. These variables capture recent healthcare-use history and allow the analysis to distinguish lower- and higher-utilization encounter segments.



Inpatient history: 0 previous visits = 8.44%; 1 previous visit = 12.92%; 2+ previous visits = 21.40%.

Emergency history shows a similar pattern: 0 previous visits = 10.47%; 1 previous visit = 14.35%; 2+ previous visits = 21.26%.

The combined analysis is even more useful because it identifies specific utilization segments rather than looking at one variable in isolation.



Strongest segment: 2+ previous inpatient visits combined with 2+ previous emergency visits has a **28.59%** observed 30-day readmission rate.

6. SQL Analysis & Diabetes-Specific Analysis

SQL was used as a complementary analytics layer. Python handled cleaning, exploratory analysis, grouping, and visualization; SQL demonstrated that the same important business metrics and segmentation could be calculated directly from the relational database. The purpose was validation and database-querying skill, not to force Python and SQL to produce different answers.

Core SQL analysis questions

- Does previous inpatient utilization indicate a higher observed 30-day readmission rate?
- How does previous emergency utilization differ across readmission outcomes?
- Which medication groups show different observed rates?
- How does hospital-stay duration relate to observed readmission?
- Which inpatient × emergency combinations have the highest observed rates?
- Which utilization segment contributes the largest share of all readmissions?
- Do A1C result, maximum serum glucose, insulin status, and diabetes medication show different observed readmission patterns?

Common SQL metric structure

```
SELECT group_column, COUNT(*) AS encounters, SUM(readmission_30d) AS readmissions_30d, ROUND(100 *  
AVG(readmission_30d), 2) AS readmission_rate FROM readmission GROUP BY group_column ORDER BY  
readmission_rate DESC;
```

Why these SQL metrics work: COUNT(*) provides the encounter denominator; SUM(readmission_30d) counts the 1s; AVG(readmission_30d) gives the proportion because the target is binary; multiplying by 100 converts it to a percentage.

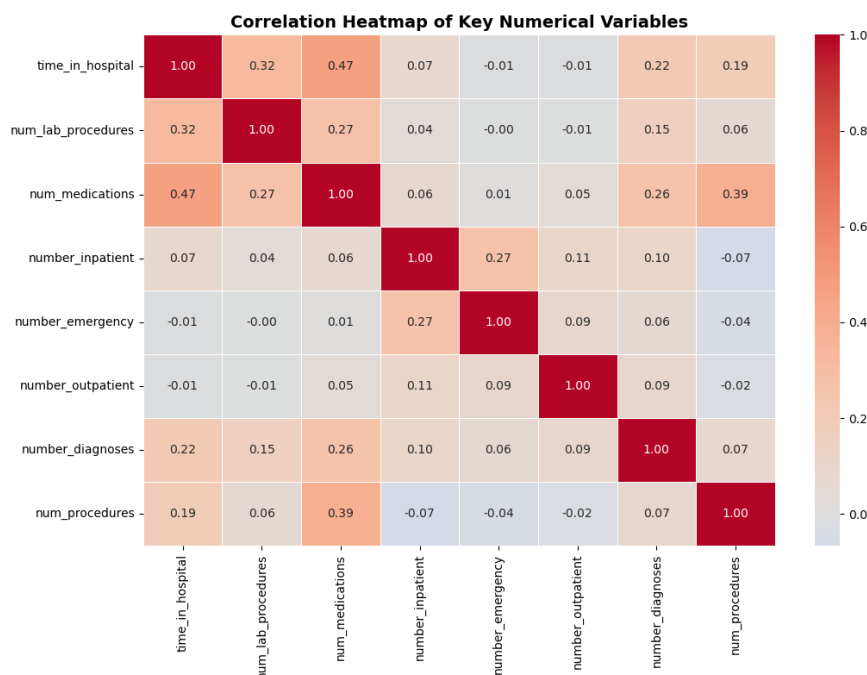
Diabetes-specific SQL analyses

A1Cresult, max_glu_serum, insulin, and diabetesMed were each analyzed with the same consistent structure: encounters, 30-day readmissions, and readmission rate. These analyses connect the utilization findings back to the diabetes context. The supplied project documentation treats them as supporting analyses rather than the headline findings.

SQL views were intentionally excluded from the final project because a reusable reporting layer was not required.

7. Correlation, Key Insights & Business Recommendations

Correlation analysis summarizes linear relationships among selected numerical variables. The strongest relationships in the uploaded matrix include time in hospital with number of medications (0.47), number of medications with number of procedures (0.39), and time in hospital with laboratory procedures (0.32). These are associations, not proof of causation.



Top Business Insights

1. **Overall baseline:** 11.16% of encounters were followed by a 30-day readmission.
2. **Prior inpatient utilization is a strong signal:** the rate rises from 8.44% with no previous inpatient visits to 21.40% with 2+ visits.
3. **Prior emergency utilization is also strong:** the rate rises from 10.47% to 21.26%.
4. **Combined utilization is the strongest segment:** 2+ inpatient + 2+ emergency reaches 28.59%.
5. **Length of stay:** longer stays generally show higher observed rates, but encounter volume declines sharply.
6. **Complexity:** higher medication and diagnosis counts show higher observed rates across the main well-populated groups.
7. **Demographics:** gender differences are small; age and race show some variation but require attention to group size.

Business Recommendations

- Use prior inpatient and emergency utilization as practical segmentation variables for identifying encounters that may warrant closer follow-up.
- Interpret readmission rates together with encounter counts so small groups do not create misleading priorities.
- Combine utilization indicators rather than relying on a single variable when identifying higher-observed-rate segments.
- Use medication and diagnosis complexity as supporting context for care-management analysis, not as causal explanations.
- For future work, add predictive modeling and time-to-event or longitudinal analysis if the goal moves from descriptive segmentation toward prediction.

Challenges & Limitations

This is an observational, descriptive analysis. The findings identify patterns and associations in the dataset; they do not establish that any factor causes readmission. Several small categories also demonstrate why percentages must be read alongside denominators.

Conclusion

The project demonstrates a complete Python + SQL analytics workflow: data-quality validation, target creation, EDA, feature grouping, multidimensional segmentation, focused visualization, database validation, and business interpretation. The clearest result is that repeated prior inpatient and emergency utilization is associated with substantially higher observed 30-day readmission rates, with the combined 2+ / 2+ segment reaching 28.59%.

The report deliberately uses focused Python visualizations rather than a chart for every column, matching the project's goal of communicating the strongest analytical findings.